
Table of Contents

--- MATH135A Project 3: Problem 2 ---	1
--- PART (a): SECANT METHOD ---	1
--- Part(b): Fixed point method ---	2

--- MATH135A Project 3: Problem 2 ---

```
% Compute the root of  $x - 0.5\sin(x) - 1 = 0$  with  $x0\_val = 0.5$  with
% Secant method with  $x1$  from one iteration of Newton's method
% Fixed point method with choice of  $g(x)$ 
format long;
% ---- Initial Setup ----
disp(' ');
disp('--- Problem 2 ---');
```

```
f2 = @(x) x - 0.5*sin(x) - 1;
df2 = @(x) 1 - 0.5*cos(x); % For Newton's Method again
true_r = 1.498701133517848;
x0 = 0.5;
```

--- Problem 2 ---

--- PART (a): SECANT METHOD ---

```
disp('Part (a): Secant Method');

% x1 using one Iteration of Newton's Method
x1 = x0 - f2(x0)/df2(x0);
fprintf('Calculated x1 (via Newton): %.10f\n', x1);

fprintf('%-5s %-15s %-20s %-15s\n', 'Iter', 'x_n', 'f(x_n)', 'Abs Error');

% Initial printing for k=0 and k=1
fprintf('%-5d %-15.10f %-20.10f %-15.10f\n', 0, x0, f2(x0), abs(x0 - true_r));
fprintf('%-5d %-15.10f %-20.10f %-15.10f\n', 1, x1, f2(x1), abs(x1 - true_r));

% Secant Loop (5 iteration)
p_prev = x0;
p_curr = x1;

for k = 2:6
    % Secant Formula
    %  $x_{new} = x_{curr} - f(x_{curr}) * (x_{curr} - x_{prev}) / (f(x_{curr}) - f(x_{prev}))$ 

    top = f2(p_curr) * (p_curr - p_prev);
    bot = f2(p_curr) - f2(p_prev);
    p_new = p_curr - (top/bot);
```

```

    % Print the current iteration results
    fprintf('%-5d %-15.10f %-20.10f %-15.10f\n', k, p_new, f2(p_new),
abs(p_new - true_r));

    % Update previous and current values for the next iteration
    p_prev = p_curr;
    p_curr = p_new;

end

```

Part (a): Secant Method

Calculated x1 (via Newton): 1.8180707002

Iter	x_n	$f(x_n)$	Abs Error
0	0.5000000000	-0.7397127693	0.9987011335
1	1.8180707002	0.3332791239	0.3193695667
2	1.4086683077	-0.0847747004	0.0900328258
3	1.4916886346	-0.0067476744	0.0070124989
4	1.4988681231	0.0001609821	0.0001669895
5	1.4987008301	-0.0000002925	0.0000003034
6	1.4987011335	-0.0000000000	0.0000000000

--- Part(b): Fixed point method ---

```

disp(' ');
disp('Part(b): Fixed Point Method');

%We will be using  $x = g(x)$ , so it will be  $g(x) = 1 + 0.5\sin(x)$ 

g = @(x) 1 + 0.5*sin(x);

x_fp = x0;
fprintf('%-5s %-15s %-15s\n', 'Iter', 'x_n', 'Abs Error');

fprintf('%-5d %-15.10f %-15.10f\n', 0, x_fp, abs(x_fp - true_r));

for k = 1:5
    x_fp = g(x_fp);
    fprintf('%-5d %-15.10f %-15.10f\n', k, x_fp, abs(x_fp - true_r));
end

disp("Does g'(x) converges or diverges?")
dg = @(x) 0.5*cos(x);
if abs(dg(true_r)) < 1
    disp(abs(dg(true_r)));
    disp('Converges');
else
    disp(abs(dg(true_r)));
    disp('Diverges');
end

```

Part(b): Fixed Point Method

<i>Iter</i>	<i>x_n</i>	<i>Abs Error</i>
0	0.5000000000	0.9987011335
1	1.2397127693	0.2589883642
2	1.4728453345	0.0258557990
3	1.4976033179	0.0010978156
4	1.4986612937	0.0000398398
5	1.4986996982	0.0000014353

Does $g'(x)$ converges or diverges?
0.036016377219443

Converges

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